

Name:Kaushal Kumar
Course:AP

Roll:CSB24069
CoureCode:CSBT215

Q: Develop an activity log analyzer in python:

You are given a list of activity records:

```
{  
    "user": str,          //roll numbers of students  
    "action": str,       //Online activities of students such as - apps, websites  
                        visited etc.  
    "duration": float    //screen time for each activity  
}
```

Soln:

```
from collections import defaultdict  
from functools import reduce  
from typing import List, Dict, Set  
  
logs = [  
    {"user": "24012", "action": "YouTube", "duration": 1.5},  
    {"user": "24013", "action": "Instagram", "duration": 2.0},  
    {"user": "24016", "action": "WhatsApp", "duration": 0.8},  
    {"user": "24014", "action": "YouTube", "duration": 1.2},  
    {"user": "24017", "action": "Chrome", "duration": 1.7},  
    {"user": "24015", "action": "Instagram", "duration": 2.5},  
    {"user": "24019", "action": "Chrome", "duration": 1.0},  
]  
  
#  
def total_time_per_user(logs: List[Dict]) -> Dict[str, float]:  
  
    result = defaultdict(float)  
  
    for log in logs:  
        result[log["user"]] += log["duration"]  
  
    return dict(result)
```

```

def most_active_users(logs: List[Dict], k: int) -> List[str]:

    totals = total_time_per_user(logs)

    return [
        user for user, _ in
        sorted(totals.items(), key=lambda x: x[1], reverse=True)[:k]
    ]

def unique_actions(logs: List[Dict]) -> Set[str]:

    return {log["action"] for log in logs}

def total_activity_time(logs: List[Dict]) -> float:

    return reduce(lambda acc, log: acc + log["duration"], logs, 0.0)

print("Total time per user:")
print(total_time_per_user(logs))

print("\nMost active users:")
print(most_active_users(logs, 2))

print("\nUnique actions:")
print(unique_actions(logs))

print("\nTotal activity time:")
print(total_activity_time(logs))

```

Output:

```
• Total time per user:  
{'24012': 1.5, '24013': 2.0, '24016': 0.8, '24014': 1.2, '24017': 1.7, '24015': 2.5, '24019': 1.0}  
  
Most active users:  
['24015', '24013']  
  
Unique actions:  
{'Instagram', 'WhatsApp', 'Chrome', 'YouTube'}  
  
Total activity time:  
10.7  
PS C:\Users\kaush\Desktop\imp\vscode\AP assignment\assignment_nr6,7,\assignment_nr6>
```

Analysis of Time complexity and space Complexity:

a) Time Complexity for computing TOP K users

1) iteration through all logs once $O(n)$

2) Sorting user by total activity time.

3) Selecting k users $O(k \log n)$
 $O(k)$

Final time Complexity

$$\begin{aligned} T(n) &= O(n + k \log n) \\ &= O(n \log n) \end{aligned}$$

b) Space Complexity of storing result

1) Dictionary storing $O(u)$

2) Sorted list of users $O(u)$

Final Space complexity $O(u)$